آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Teaching and Learning Resources for Grade IX Chemistry

Recommended Key Textbook:

Chemistry for Grade 9, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Additional Recommended Resource Material				
Topic No.	**Topic**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
1	Fundamentals of Chemistry	*Chemistry 9,* Caravan Book House, Lahore.	Unit 1: Fundamentals of Chemistry	Atoms & Ions: https://www.youtube.com/watch?v=fN8kH9Vvqo0&list=RDCMUCaGEe4KXZrjou9kQx6ezG2w&index=7
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level,* Marshall Cavendish Education.	Chapter 4: Element, compound and mixtures	

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2	Stoichiometry	Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level,* Marshall Cavendish Education.	Chapter 8: Writing Equation	https://phet.colorado.edu/en/simulations/balancing-chemical-equations
		Chemistry for Class 9, Punjab Textbook Board, Lahore.	Chapter 2: Chemical Combination and Chemical Equations	Chemical Reactions and Equations: https://www.youtube.com/watch?v=kYTCruQZdaU
		Christopher N. Prescott, *Chemistry, A Course for 'O' Level,* Marshall Cavendish Education (Third Edition).	Unit 7: Chemical Formulae and Equations Unit 8: The Mole	
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 4: Reacting Masses and Chemical Equations Unit 5: Using Moles	
3	Atomic Structure	Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level,* Marshall Cavendish Education.	Chapter 5: Atomic Structure	https://phet.colorado.edu/sims/html/build-an-atom/latest/build-an-atom_en.html
		Chemistry 9, Caravan Book House, Lahore.	Unit 2: Structure of Atoms	https://www.britannica.com/science/atomic-radius
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 2: Atoms and Elements	

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4	Periodic Table and Periodicity	*Chemistry 9*, Caravan Book House, Lahore.	Unit 3: Periodic Table and Periodicity of Properties	Perfecting the Periodic Table: https://www.youtube.com/watch?v=7tbMGKGgCRA https://www.britannica.com/science/periodic-table/The-first-periodic-table Periodic Trends: https://www.youtube.com/watch?v=7cEtOHLZQ2A
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 12: The Periodic Table	
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Unit 16: The Periodic Table	
5	Structure of Molecules	Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 3: Atoms Combining	Chemical Bonding: https://www.youtube.com/watch?v=1g_psVD4bl0&list=PLE12331E9AB4BAE14 http://www.ausetute.com.au/intermof.html https://www.bbc.co.uk/bitesize/guides/zjfkw6f/revision/5
		Chemistry for Class IX, Sindh Textbook Board, Jamshoro.	Chapter 4: Chemical Bonding	
		Chemistry 9, Caravan Book House, Lahore.	Unit 4: Structure of Molecules	
		Dr Rex M. Heyworth and J. G. R. Briggs, *All About Chemistry 'O' Level*, Hodder Education Singapore.	Chapter 6: Chemical Bonds Chapter 7: Structure and Properties	
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 7: Covalent and Metallic Bonding	

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6	States of Matter	*Chemistry 9*, Caravan Book House, Lahore.	Unit 5: Physical States of Matter	https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Physical_Properties_of_Matter/States_of_Matter/Properties_of_Gases/Gas_Laws/Gas_Laws_-_Overview#:~:text=Boyle's%20Law%20tells%20us%20that,the%20amount%20of%20gas%20increases.
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 1: States of Matter Unit 3: Atoms Combining	https://www.bbc.co.uk/bitesize/guides/zjfkw6f/revision/4
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 7: Covalent and Metallic Bonding	
		Christopher N. Prescott, *Chemistry, A Course for 'O' Level*, Marshall Cavendish Education (Third Edition).	Unit 2: Particles of Matter	

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7	Solutions	*Chemistry 9*, Caravan Book House, Lahore.	Unit 6: Solutions	http://www.chemguide.co.uk/analysis/chromatography/paper.html http://www.bbc.co.uk/schools/gcsebitesize/science/add_edexcel/covalent_compounds/seperationrev2.shtml
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 3: Purification and Separation	
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 18: Separation and |Purification	
		Chemistry for Class IX, Sindh Textbook Board, Jamshoro.	Chapter 6: Solutions	
8	Electrochemistry	*Chemistry 9*, Caravan Book House, Lahore.	Unit 7: Electrochemistry	Electrolysis: https://www.youtube.com/watch?v=RAEm-kJ_QkU https://www.onlinemathlearning.com/reactivity-series.html Corrosion: https://www.youtube.com/watch?v=TKMgUCq3npg
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 14: Metals Chapter 15: Electrolysis	
		Christopher N. Prescott, *Chemistry, A Course for 'O' Level*, Marshall Cavendish Education (Third Edition).	Unit 9: Electricity and Chemicals Unit 12: Redox Reactions Unit 18: Metals	

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Note: This resource list has been prepared primarily for teachers. While it can be shared with students, students should not be required to buy multiple books. Schools are encouraged to stock these books in the library. Moreover, these are only suggestions which have been compiled for the ease of teachers and students; schools are encouraged to use other resources for teaching and learning as well, as long as they are in line with the Student Learning Outcomes (SLOs) mentioned in AKU-EB syllabi. Unless specified, AKU-EB does not endorse any of these books or websites. You are advised to use an ad-blocker while accessing the websites. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine. If you have any query, please contact us via email.

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Teaching and Learning Resources for Grade X Chemistry

Recommended Key Textbook:

Chemistry for Grade 10, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Additional Recommended Resource Material				
Topic No.	**Topic**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
9	Chemical Equilibrium	*Chemistry 10*, Caravan Book House, Lahore.	Unit 9: Chemical Equilibrium	http://www.ausetute.com.au/lechatsp.html
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 17: Energy Changes	http://www.chemguide.co.uk/physical/equilibria/lechatelier.html Exothermic and Endothermic Reactions:
		Christopher N. Prescott, *Chemistry, A Course for 'O' Level*, Marshall Cavendish Education (Third Edition).	Unit 10: Energy from Chemicals	https://www.youtube.com/watch?v=dstRL5xB0Sk
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 8: Energy Changes in Reactions Unit 10: Reversible reactions and equilibrium	

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Topic No.	Topic	Reference Book	Topic in the Reference Book	Website/ Link
		Chemistry for Class 11. Punjab Textbook Board, Lahore.	Chapter 8: Chemical Equilibrium	
10	Acids, Bases and Salts	*Chemistry 10*, Caravan Book House, Lahore.	Unit 10: Acids, Bases and Salts	
		Chemistry for Class X, Sindh Textbook Board, Jamshoro.	Chapter 2: Acid, Base and Salt	
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 11: Acids and Bases Chapter 12: Salts	
11	Organic Chemistry	*Chemistry 10*, Caravan Book House, Lahore.	Unit 11: Organic Chemistry	The Functional Group: https://www.youtube.com/watch?v=nMTQKBn2Iss https://www.teachoo.com/17724/3507/What-are-isomers-/category/Concepts/
		Chemistry for Class XII, Sindh Textbook Board, Jamshoro.	Chapter 3: Organic Compounds Chapter 4: Nomenclature of Organic Compounds	
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 16: Organic Chemistry	

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Topic No.	Topic	Reference Book	Topic in the Reference Book	Website/ Link
12	Hydrocarbons	*Chemistry 10*, Caravan Book House, Lahore.	Unit 12: Hydrocarbons	https://www.chem.ucalgary.ca/courses/351/WebContent/orgnom/index.html
		Chemistry for Class XII, Sindh Textbook Board, Jamshoro.	Chapter 5: Hydrocarbons	
		Chemistry for Class 10, Punjab Textbook Board, Lahore.	Chapter 17: Organic Chemistry	
		Dr Rex M. Heyworth and J. G. R. Briggs, *All About Chemistry 'O' Level*, Hodder Education Singapore.	Chapter 25: Alkanes and Alkenes	
13	Natural and Synthetic Polymers	*Chemistry 10*, Caravan Book House, Lahore.	Unit 13: Biochemistry	Protein Structure and Protein Denaturation: https://www.youtube.com/watch?v=O5uqdxQyJj8
		Chemistry for Grade 12, National Book Foundation, Islamabad.	Chapter 21: Biochemistry	http://glencoe.mheducation.com/sites/9834092339/student_view0/chapter3/protein_denaturation.html
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 17: Polymers	https://www.britannica.com/science/agricultural-sciences https://www.britannica.com/science/nutrition

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Topic No.	Topic	Reference Book	Topic in the Reference Book	Website/ Link
14	Environmental Chemistry I: Atmosphere	*Chemistry 10*, Caravan Book House, Lahore.	Unit 14: The Atmosphere	https://gml.noaa.gov/outreach/info_activities/pdfs/TBI_earth_spheres.pdf
		Chemistry for Grade 12, National Book Foundation, Islamabad.	Chapter 23: Environmental Chemistry	
		Dr Rex M. Heyworth and J. G. R. Briggs, *All About Chemistry 'O' Level*, Hodder Education Singapore.	Chapter 23: Air and the Environment	
15	Environmental Chemistry II: Water	*Chemistry 10*, Caravan Book House, Lahore.	Unit 15: Water	Waste Water Treatment: https://www.youtube.com/watch?v=jLaeBykDwaM
		Chemistry for Class 10, Punjab Textbook Board, Lahore.	Chapter 11: Hydrogen and Water	https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Chemistry_for_Changing_Times_(Hill_and_McCreary)/14%3A_Water/14.01%3A_Water_-_Some_Unique_Properties
		Chemistry for Class X, Sindh Textbook Board, Jamshoro.	Chapter 6: Water	
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 15: Chemistry of the Environment	

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16	Chemical Industries	*Chemistry 10*, Caravan Book House, Lahore.	Unit 16: Chemical Industries	http://www.chemguide.co.uk/inorganic/extraction/aluminium.html
		Christopher N. Prescott, *Chemistry, A Course for 'O' Level*, Marshall Cavendish Education (Third Edition).	Unit 15: Ammonia and its Uses Unit 16: Sulphuric acid Unit 19: Extraction of Metals	http://www.chemguide.co.uk/physical/equilibria/contact.html http://www.chemguide.co.uk/physical/equilibria/haber.html
		Rose Marie Gallagher and Paul Ingram, *Complete Chemistry*, Oxford University Press (Fourth Edition).	Unit 14: Extracting and Using Metals	https://www.bbc.co.uk/bitesize/guides/zcc6dxs/revision/1
		Tan Yin Toon, Chen Ling Kwong, John Sadler and Emily Clare, *Chemistry Matters G.C.E. 'O' level*, Marshall Cavendish Education.	Chapter 19: Ammonia	

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